Task	Prompt	Answer
weighted minimum spanning tree	The task is to determine the minimum spanning tree of a weighted graph. A minimum spanning tree is a subset of the edges that connects all vertices in the graph with the minimum possible total edge weights. If not specified, all edges have equal edge weights. The input graph is guaranteed to be undirected and connected. Here is an undirected graph containing nodes from 1 to 5. The edges are: (1, 4, 5), (2, 4, 11), (2, 3, 10), (3, 4, 2), (3, 5, 2). (u ,v, w) denotes the edge from node *u* to node *v* has a weight of *w*. Question: What is the minimum spanning tree of the weighted graph? You need to format your answer as a list of edges in ascending dictionary order, e.g., [(u1, v1), (u2, v2), ..., (un, vn)].	[(1, 4), (2, 3), (3, 4), (3, 5)]
weighted shortest path	The task is to determine the shortest path between two nodes of a weighted graph. The input nodes are guaranteed to be connected. Here is a directed graph containing nodes from 1 to 8. The edges are: (1, 2, 5), (1, 4, 3), (1, 7, 9), (2, 3, 10), (2, 4, 10), (3, 1, 11), (3, 4, 2), (3, 5, 6), (4, 1, 1), (4, 2, 4), (4, 6, 8), (4, 8, 2), (5, 1, 7), (5, 2, 11), (5, 6, 2), (5, 7, 5), (5, 8, 11), (6, 1, 7), (6, 2, 11), (6, 3, 4), (6, 5, 1), (6, 8, 11), (7, 1, 3), (7, 2, 8), (7, 4, 7), (7, 6, 6), (7, 8, 3), (8, 1, 11), (8, 2, 7), (8, 4, 5), (8, 7, 5). (u ,v, w) denotes the edge from node *u* to node *v* has a weight of *w*. Question: What is the shortest path between node 1 and node 5? You need to format your answer as a list of nodes, e.g., [node-1, node-2, ..., node-n].	[1, 4, 6, 5]
wiener index	The task is to determine the Wiener index of a connected graph. The Wiener index of a graph is the sum of the shortest-path distances between each pair of reachable nodes. For pairs of nodes in undirected graphs, only one orientation of the pair is counted. In the input graph, all node pairs are guaranteed to be reachable. Here is an undirected graph containing nodes from 1 to 5. The edges are: (1, 2), (1, 4), (2, 3), (4, 5), (3, 5). Question: What is the Wiener index of the graph? You need to format your answer as a float number.	15.0